

One Difference, Not Two Separate Intervals

AP Statistics · Unit 3 · Topic 3.10 · B: 2026-12-11 · E: 2027-01-22

[STAR] TODAY'S PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: “The two-sample z -interval is $(0.030, 0.270)$.”

Devon: “Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible.”

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE — before the video (5 min)

Which method gives the requested 95% interval for $p_O - p_P$? Cite one design or numerical detail.

[BOOK] BUILD ONE INTERVAL FOR THE DIFFERENCE (keep on the page)

For two independent random samples, estimate $p_1 - p_2$ with a two-sample z -interval: $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\hat{p}_1(1 - \hat{p}_1)/n_1 + \hat{p}_2(1 - \hat{p}_2)/n_2}$. Verify independent random samples or random assignment, $n_i \leq 0.10N_i$ when sampling without replacement, and at least 10 observed successes and failures in each group. For an interval, use each group's sample proportion separately; because the samples are independent, estimated variances add before taking the square root. Two separately constructed $C\%$ intervals do not automatically combine into one $C\%$ interval for $p_1 - p_2$.

Finish after the video:

DOK 1 (a) Define p_O and p_P and name the interval procedure for $p_O - p_P$.

DOK 2 (b) Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.

DOK 3 (c) Choose the warranted method. Use design, counts, variance structure, and zero's location. Explain why separate 95% intervals do not produce the requested coverage and how Devon's result misleads.

Frame: The warranted method is _____ because the data come from _____, and the four counts are _____.

Frame: Devon's interval misleads a reader because _____; as a result, zero appears _____ even though _____.

EXIT — turn this in

One thing the video changed about my first take: _____